these two regions will be separated by a small surface. The size of this surface is its *area*. The second quantity, after the volume, which characterizes the quantum webs of space, is the area associated with each line.[fn35]

The area, just as in the case of the volume, is a physical variable, and has a spectrum which may be calculated using Dirac's equation.[*] Area is not continuous, it is granular. There is no such thing as an arbitrarily small area.

Space appears continuous to us only because we cannot perceive the extremely small scale of these individual quanta of space. Just as when we look closely at the cloth of a T-shirt, we see that it is woven from small threads.

When we say that the volume of a room is, for example, 100 cubic metres, we are in effect *counting* the grains of space – the quanta of the gravitational field – which it contains. In a room, this number has more than a hundred digits. When we say that the area of this page is 200 square centimetres, we are actually counting the number of links in the web, or loops, which traverse the page. Across the page of this book, there is a number of quanta with more or less seventy digits.

The idea that measuring length, area and volume is a question of counting individual elements had been proposed in the nineteenth century by Riemann himself. The mathematician who had developed the theory of *continuous* curved mathematical spaces was already aware that a *discrete* physical space is, ultimately, more reasonable than a continuous one.

To summarize, the theory of loop quantum gravity, or loop theory, combines general relativity with quantum mechanics in a rather conservative way, because it does not employ any other hypothesis apart from those of the two theories themselves, suitably rewritten to render them compatible. But the consequences are radical.

General relativity taught us that space is something dynamic, like the electromagnetic field: an immense, mobile mollusc in which we are immersed, which stretches and bends. Quantum mechanics teaches us that every field of this sort is made of quanta, that is to say, it has a fine, granular structure. It follows that physical space, being a field, is made of quanta as well. The same granular structure characterizing the other quantum fields also characterizes the quantum gravitational field, and therefore space. We expect space to be granular. We expect quanta of gravity, just as there are quanta of light, quanta of the electromagnetic field, and as particles are quanta of quantum fields. But space is the gravitational field, and the quanta of the gravitational field are quanta of space: the granular constituents of space.

The central prediction of loop theory is therefore that space is not a continuum, it is not divisible ad infinitum, it is formed of 'atoms of space'. A billion billion times smaller than the smallest of atomic nuclei.